Introduction to Machine Learning
Week 1 – March 1

Course: Machine Learning

Programme: MSc

Lecturer :

Sewwandi Mahaliyana

MSc In Computing

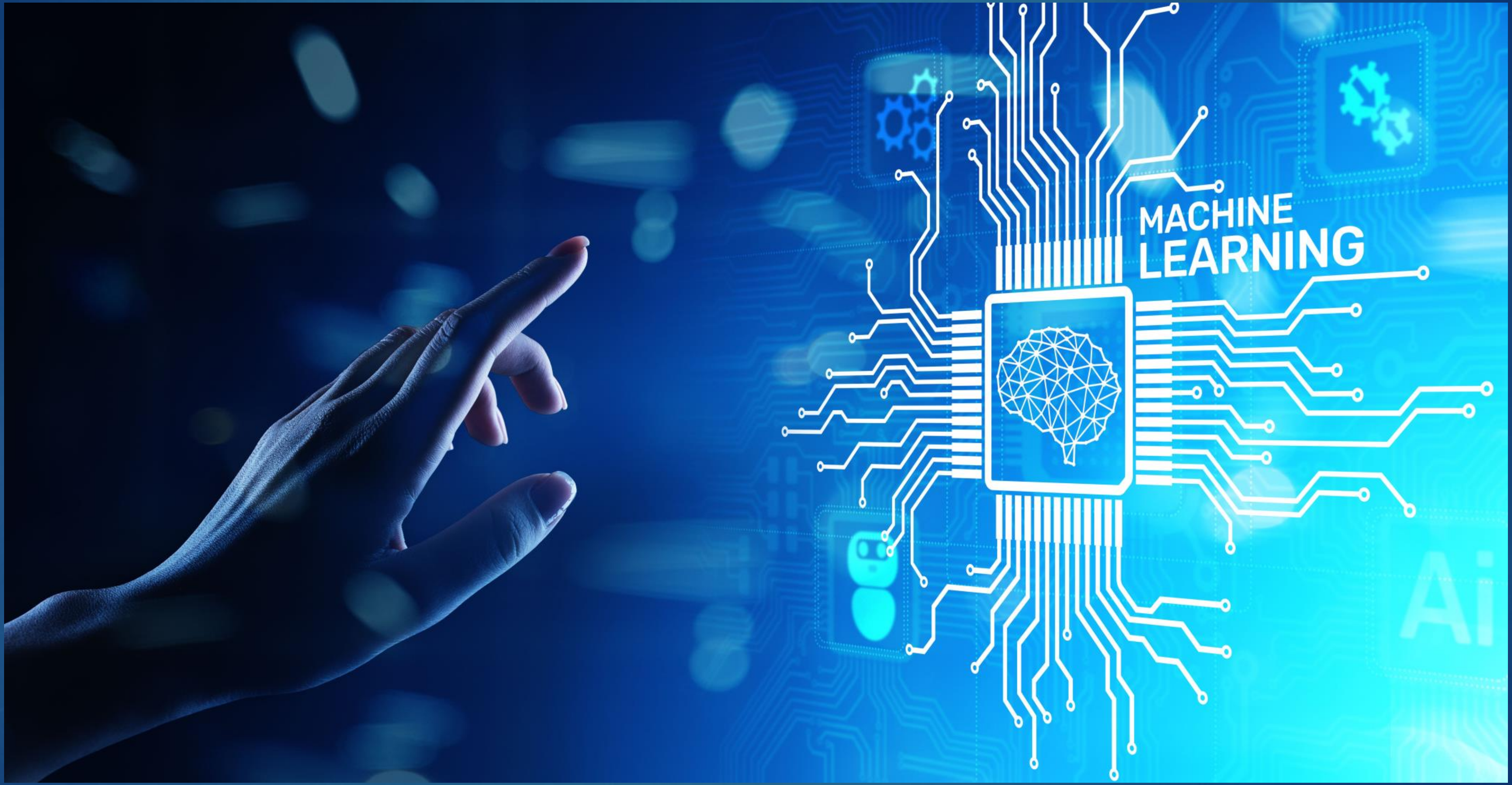
BSc(Hons)computing

HNDIT

What is Machine Learning?

- Branch of Artificial Intelligence (AI)
- Systems learn from data
- No need to explicitly program every rule
- Improves automatically with experience

- Definition: ML enables systems to learn patterns from data and make predictions.

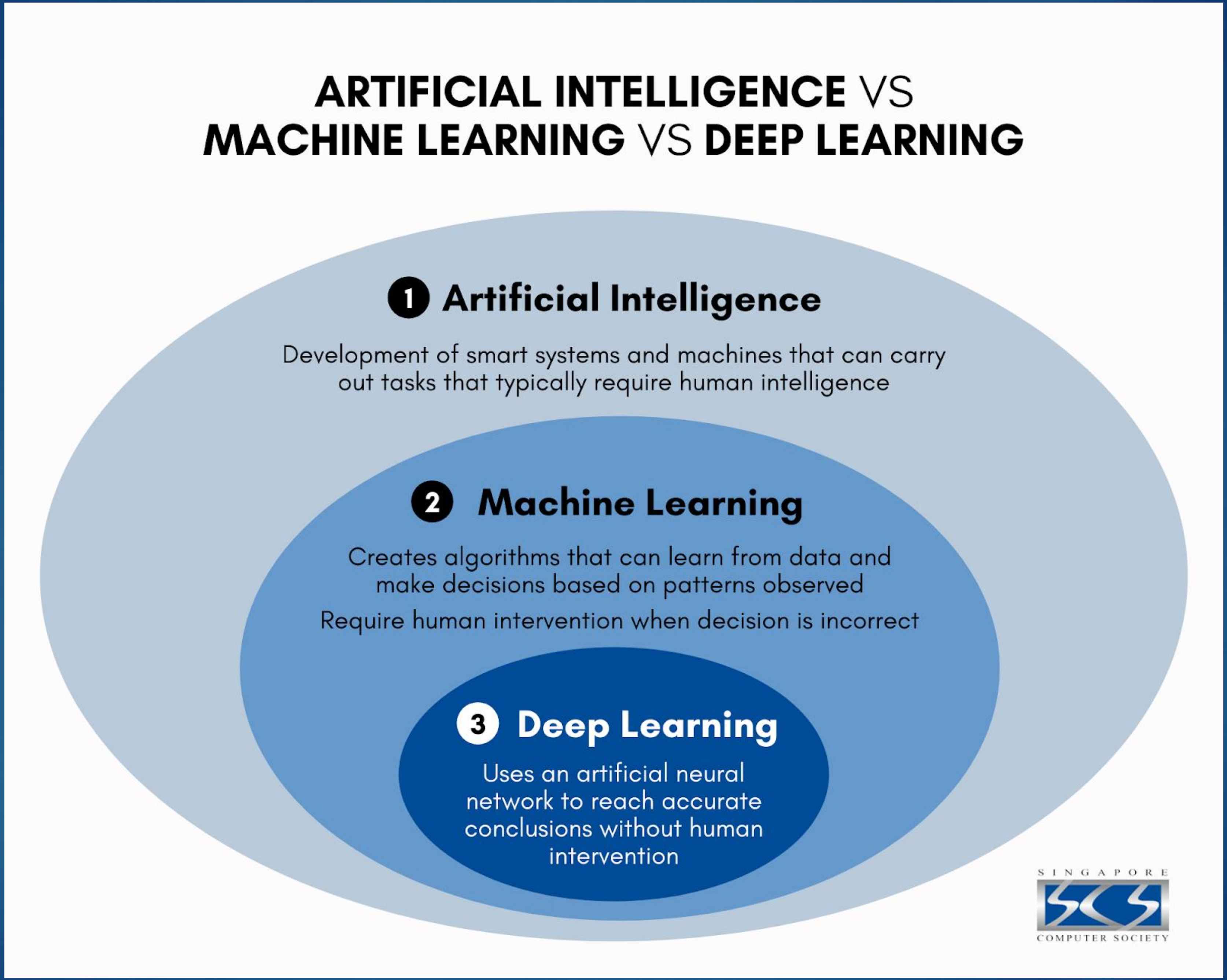

MACHINE
LEARNING
Ai

Why Machine Learning is Important?

- Handles large amounts of data
- Automates decision-making
- Improves accuracy over time
- Reduces human effort
- Used in modern intelligent systems

AI vs ML vs Deep Learning

- • Artificial Intelligence (AI) – Broad concept
- • Machine Learning (ML) – Subset of AI
- • Deep Learning (DL) – Subset of ML
- Examples:
  - ML – Email spam filtering
  - DL – Image recognition


ARTIFICIAL INTELLIGENCE VS
MACHINE LEARNING VS DEEP LEARNING
1 Artificial Intelligence
Development of smart systems and machines that can carry out tasks that typically require human intelligence
2 Machine Learning
Creates algorithms that can learn from data and make decisions based on patterns observed
Require human intervention when decision is incorrect
3 Deep Learning
Uses an artificial neural network to reach accurate conclusions without human intervention
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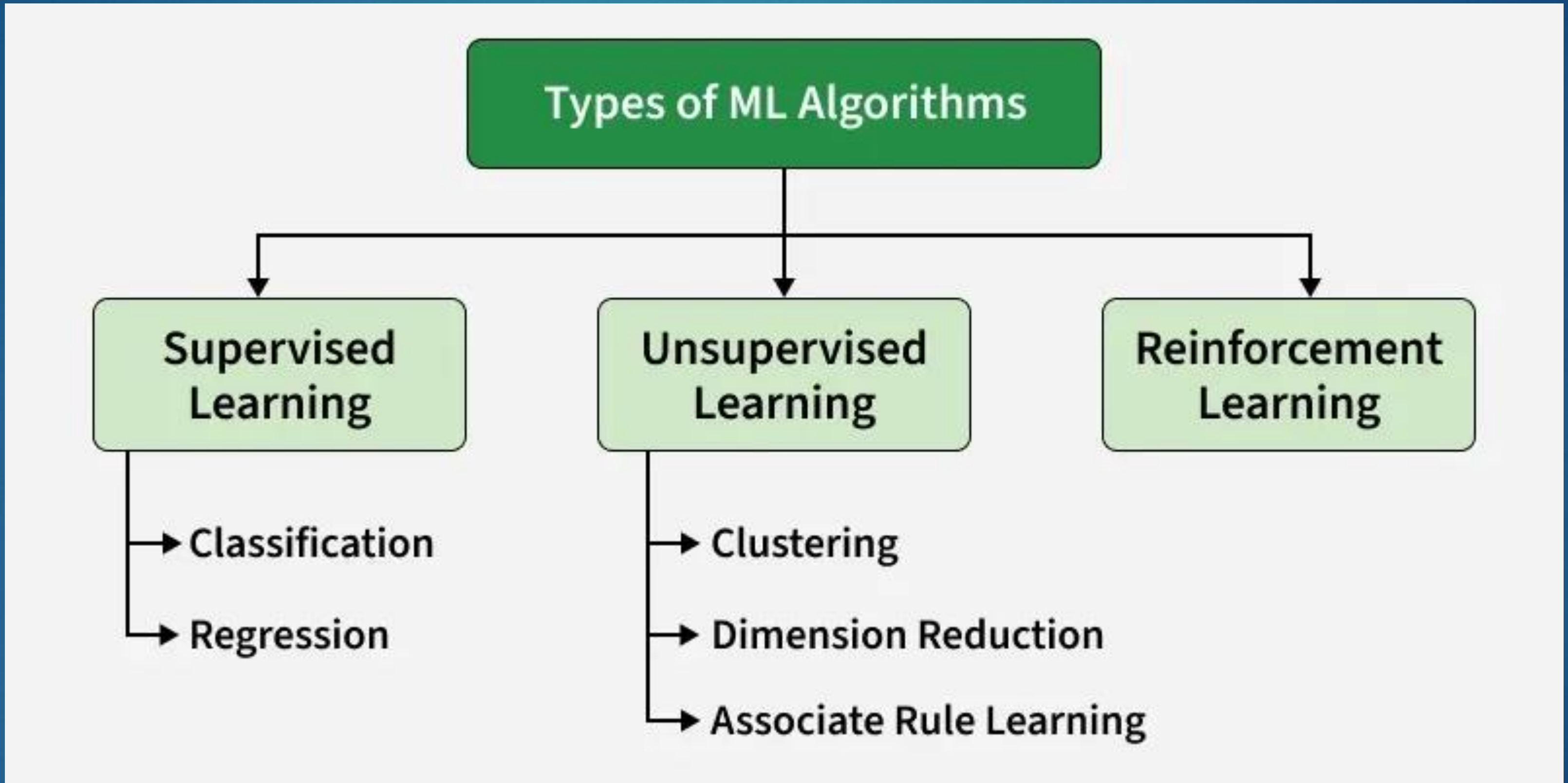
Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

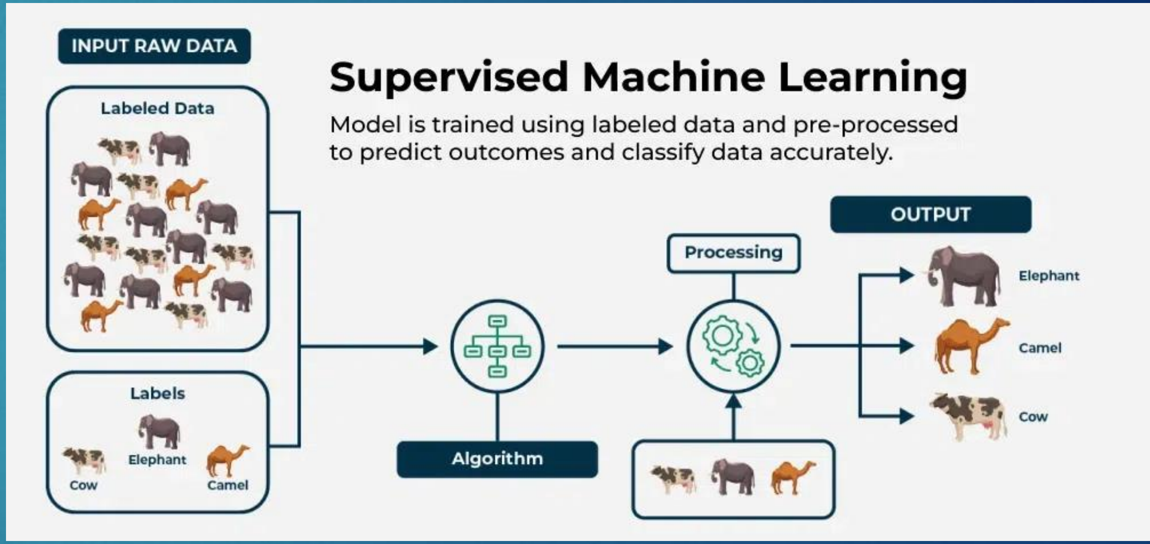
- Classification is based on how the model learns from data.



Types of ML Algorithms
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Classification
Regression
Clustering
Dimension Reduction
Associate Rule Learning

Supervised Learning

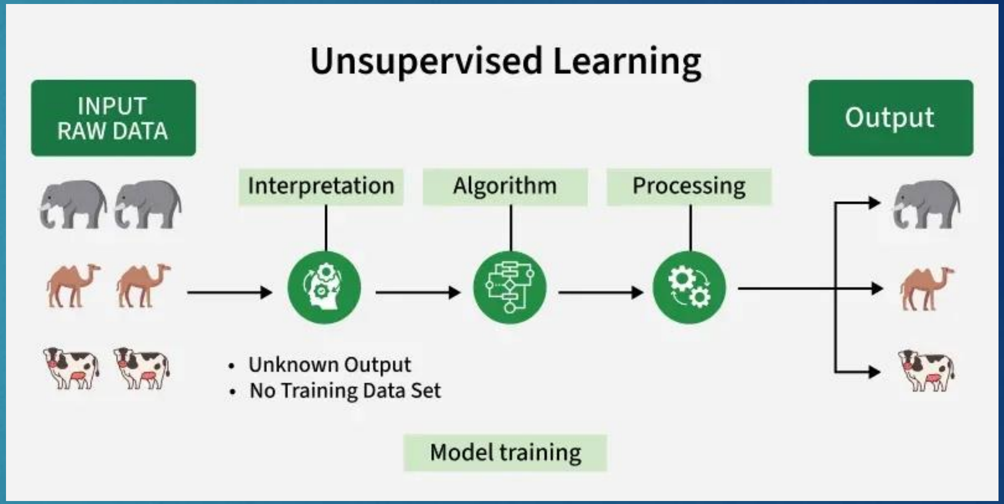
- Uses labeled data
- Input + Correct Output provided
- Learns mapping function



- Examples:
  - Email spam detection
  - House price prediction
  - Student performance prediction

Unsupervised Learning

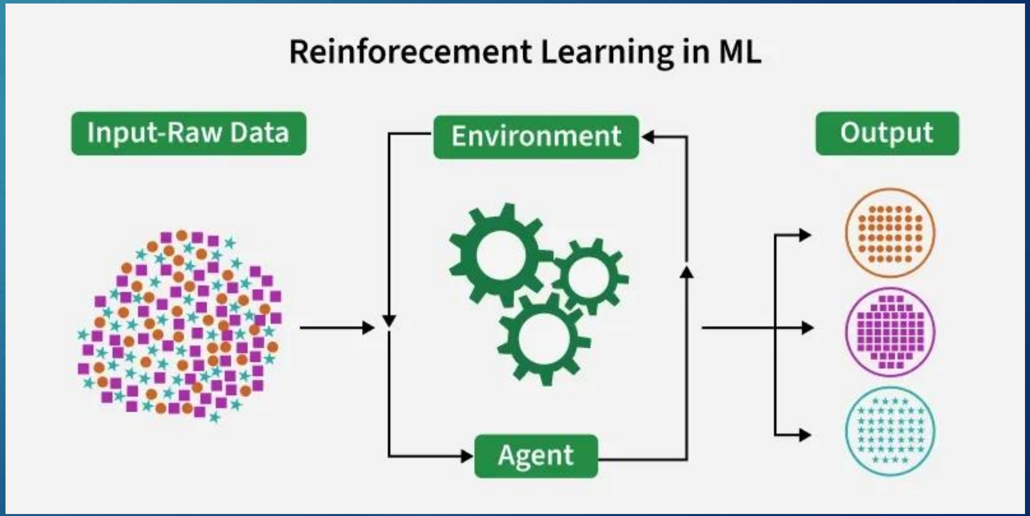
- Uses unlabeled data
- Finds hidden patterns
- No predefined output



- Examples:
  - Customer segmentation
  - Market basket analysis

Reinforcement Learning

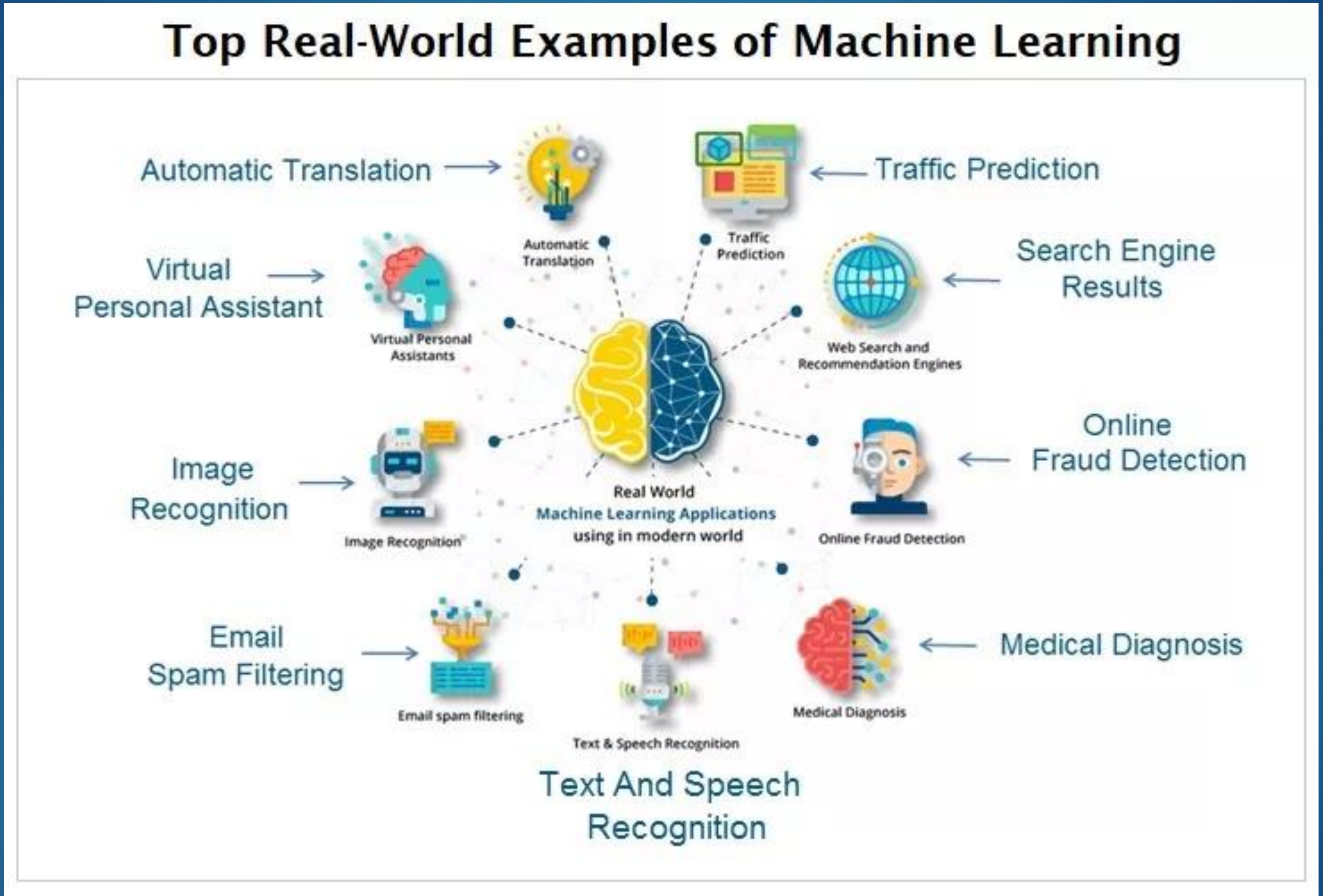
- Learning through trial and error
- Agent interacts with environment
- Reward and punishment system



- Examples:
  - Self-driving cars
  - Game playing (Chess, Go)

Real-World Applications of ML

- Healthcare – Disease prediction
- Finance – Fraud detection
- Education – Performance prediction
- Social Media – Recommendation systems
- Retail – Customer behavior analysis



Top Real-World Examples of Machine Learning
Automatic Translation
Traffic Prediction
Virtual Personal Assistant
Search Engine Results
Image Recognition
Online Fraud Detection
Email Spam Filtering
Medical Diagnosis
Text And Speech Recognition
Automatic Translation
Traffic Prediction
Virtual Personal Assistants
Web Search and Recommendation Engines
Real World
Machine Learning Applications
using in modern world
Image Recognition
Online Fraud Detection
Email spam filtering
Medical Diagnosis
Text & Speech Recognition

Simple Dataset Demonstration

- Example: Student Exam Results

Study Hours	Attendance	Result
5 hours	80%	Pass
2 hours	50%	Fail
7 hours	90%	Pass

- Type: Supervised Learning

R15

	EEID	Full Name	Job Title	Department	Business Unit	Gender	Ethnicity	Age	Hire Date	Annual Salar	Bonus %	Country	City	Exit Date
2	E02387	Emily Davis	Sr. Manger	IT	Research & Development	Female	Black	55	4/8/2016	$141,604	15%	United States	Seattle	10/16/2021
3	E04105	Theodore Dinh	Technical Architect	IT	Manufacturing	Male	Asian	59	11/29/1997	$99,975	0%	China	Chongqing	
4	E02572	Luna Sanders	Director	Finance	Speciality Products	Female	Caucasian	50	10/26/2006	$163,099	20%	United States	Chicago	
5	E02832	Penelope Jordan	Computer Systems Manager	IT	Manufacturing	Female	Caucasian	26	9/27/2019	$84,913	7%	United States	Chicago	
6	E01639	Austin Vo	Sr. Analyst	Finance	Manufacturing	Male	Asian	55	11/20/1995	$95,409	0%	United States	Phoenix	
7	E00644	Joshua Gupta	Account Representative	Sales	Corporate	Male	Asian	57	1/24/2017	$50,994	0%	China	Chongqing	
8	E01550	Ruby Barnes	Manager	IT	Corporate	Female	Caucasian	27	7/1/2020	$119,746	10%	United States	Phoenix	
9	E04332	Luke Martin	Analyst	Finance	Manufacturing	Male	Black	25	5/16/2020	$41,336	0%	United States	Miami	5/20/2021
10	E04533	Easton Bailey	Manager	Accounting	Manufacturing	Male	Caucasian	29	1/25/2019	$113,527	6%	United States	Austin	
11	E03838	Madeline Walker	Sr. Analyst	Finance	Speciality Products	Female	Caucasian	34	6/13/2018	$77,203	0%	United States	Chicago	
12	E00591	Savannah Ali	Sr. Manger	Human Resources	Manufacturing	Female	Asian	36	2/11/2009	$157,333	15%	United States	Miami	
13	E03344	Camila Rogers	Controls Engineer	Engineering	Speciality Products	Female	Caucasian	27	10/21/2021	$109,851	0%	United States	Seattle	
14	E00530	Eli Jones	Manager	Human Resources	Manufacturing	Male	Caucasian	59	3/14/1999	$105,086	9%	United States	Austin	
15	E04239	Everleigh Ng	Sr. Manger	Finance	Research & Development	Female	Asian	51	6/10/2021	$146,742	10%	China	Shanghai	
16	E03496	Robert Yang	Sr. Analyst	Accounting	Speciality Products	Male	Asian	31	11/4/2017	$97,078	0%	United States	Austin	3/9/2020
17	E00549	Isabella Xi	Vice President	Marketing	Research & Development	Female	Asian	41	3/13/2013	$249,270	30%	United States	Seattle	
18	E00163	Bella Powell	Director	Finance	Research & Development	Female	Black	65	3/4/2002	$175,837	20%	United States	Phoenix	
19	E00884	Camila Silva	Sr. Manger	Marketing	Speciality Products	Female	Latino	64	12/1/2003	$154,828	13%	United States	Seattle	
20	E04116	David Barnes	Director	IT	Corporate	Male	Caucasian	64	11/3/2013	$186,503	24%	United States	Columbus	
21	E04625	Adam Dang	Director	Sales	Research & Development	Male	Asian	45	7/9/2002	$166,331	18%	China	Chongqing	
22	E03680	Elias Alvarado	Sr. Manger	IT	Manufacturing	Male	Latino	56	1/9/2012	$146,140	10%	Brazil	Manaus	
23	E04732	Eva Rivera	Director	Sales	Manufacturing	Female	Latino	36	4/2/2021	$151,703	21%	United States	Miami	
24	E03484	Logan Rivera	Director	IT	Research & Development	Male	Latino	59	5/24/2002	$172,787	28%	Brazil	Rio de Janerio	
25	E00671	Leonardo Dixon	Analyst	Sales	Speciality Products	Male	Caucasian	37	9/5/2019	$49,998	0%	United States	Seattle	
26	E02071	Mateo Her	Vice President	Sales	Speciality Products	Male	Asian	44	3/2/2014	$207,172	31%	China	Chongqing	
27	E02206	Jose Henderson	Director	Human Resources	Speciality Products	Male	Black	41	4/17/2015	$152,239	23%	United States	Columbus	
28	E04545	Abigail Mejia	Quality Engineer	Engineering	Corporate	Female	Latino	56	2/5/2005	$98,581	0%	Brazil	Rio de Janerio	
29	E00154	Wyatt Chin	Vice President	Engineering	Speciality Products	Male	Asian	43	6/7/2004	$246,231	31%	United States	Seattle	
30	E03343	Carson Lu	Engineering Manager	Engineering	Speciality Products	Male	Asian	64	12/4/1996	$99,354	12%	China	Beijing	
31	E00304	Dylan Choi	Vice President	IT	Corporate	Male	Asian	63	5/11/2012	$231,141	34%	China	Beijing	
32	E02594	Ezekiel Kumar	IT Coordinator	IT	Research & Development	Male	Asian	28	6/25/2017	$54,775	0%	United States	Columbus	

Learning Outcomes

- After this session, students should be able to:
  - Understand ML concepts
  - Differentiate ML types
  - Identify ML applications
  - Recognize simple datasets

Class Activities

- Activity 1: Discuss real-life ML examples
- Activity 2: Identify inputs and outputs in dataset
- Activity 3: Decide which ML type fits the problem